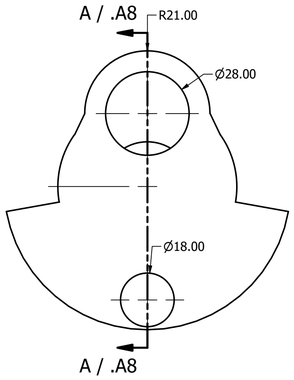
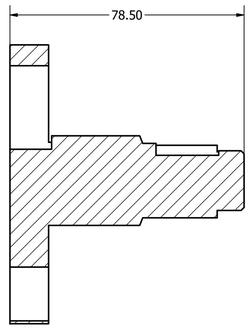
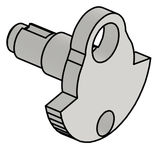
TCE-101 CRANKWEBLEFT 4340 QT, QTY 1, 746.6 g (INVENTOR), SCALE 1:1, ENVELOPE 94.5 x 93.5 x 78.5 MM
WEB 13: HUB R30, PIN BOSS R21, COUNTERWEIGHT R48 x 160 DEG, W-ALLOY SLUG BORE 18 AT r 38; CRANKPIN BORE 27.915 +/-0.004 AT r 24.50 +/-0.01;
30 k5 JOURNAL (6306 + 30x47x7 SEAL), 25 h6 FLYWHEEL SEAT + ROUND KEY 8, M20x1.5 THREAD

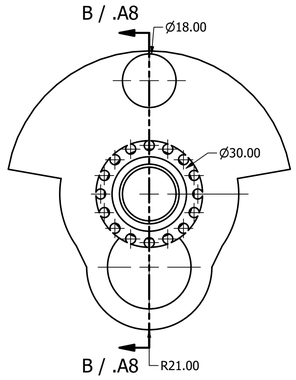


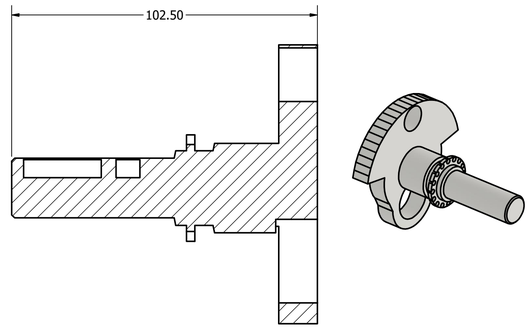


SECTION A-A
SCALE 1 : 1 / .D8



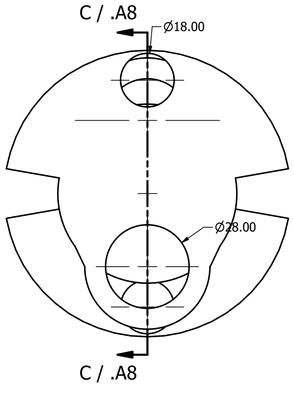
TCE-102 CRANKWEBRIGHT 4340 QT, 16T TEETH IND. HARDENED, QTY 1, 764.3 g (INVENTOR), SCALE 1:1, ENVELOPE 102.5 x 94.5 x 93.5 MM
AS LEFT WEB; 30 k5 JOURNAL, INTEGRAL 16T ANSI 25 CAM-CHAIN SPROCKET (PD 32.55, TEETH 2.8 WIDE),
20 h6 PUMP / SEAL / OUTPUT JOURNAL, PUMP DRIVE SLOT 6, KEYWAY 6 (DYNO COUPLING)

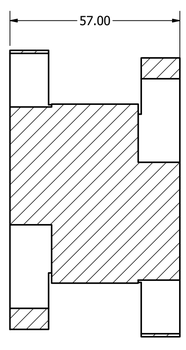




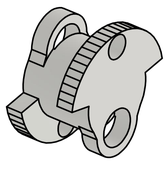
SECTION B-B
SCALE 1 : 1 / .D4

TCE-103 CRANKCENTER 4340 QT, QTY 1, 1598.4 g (INVENTOR), SCALE 1:1, ENVELOPE 96.0 x 94.5 x 57.0 MM
ONE PIECE: TWO WEBS 13 AT 180 DEG ON AN R30 HUB (31 WIDE), 2 CRANKPIN BORES 27.915 +/-0.004 AT r 24.50 +/-0.01, 180 DEG +/-0.05,
2 SLUG BORES 18 AT r 38; NO CENTRE MAIN (TWO-MAIN CRANK)

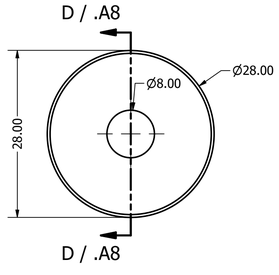


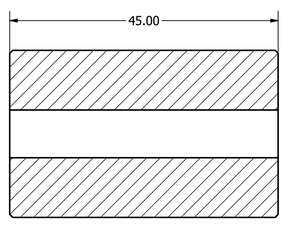


SECTION C-C
SCALE 1 : 1 / .B8

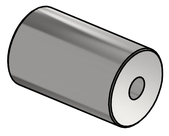


TCE-104 CRANKPIN 52100, 60 HRC, GROUND, QTY 2, 198.6 g (INVENTOR), SCALE 2:1, ENVELOPE 45.0 x 28.0 x 28.0 MM
28 g5 x 45, 52100 60 HRC, Ra 0.2 (K28x34x17 NEEDLE RACE), BORE 8; PRESSED INTO 27.915 +/-0.004 WEB BORES (0.065-0.082) (QTY 2)





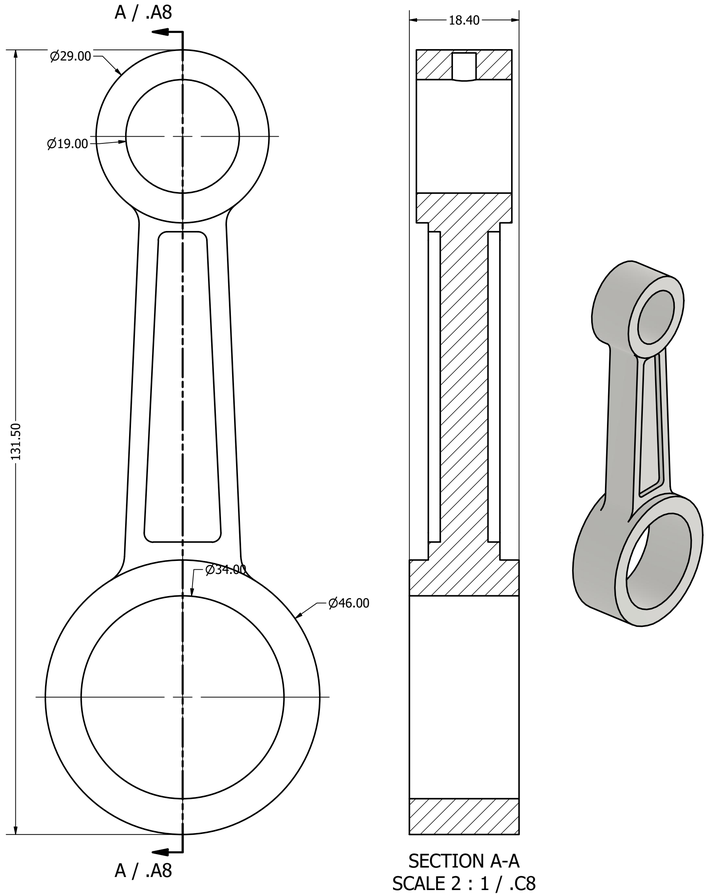
SECTION D-D
SCALE 2 : 1 / .B4



NOTES:
1. 4340 QT 36-40 HRC (CRANK WEBS, CENTRE, RODS); 52100 60 HRC GROUND (CRANKPINS); 6061-T6 BILLET FOR THE ONE-OFF CASE, BLOCK, HEAD
   (PRODUCTION: A356-T6 SAND CASTING). BREAK SHARP EDGES 0.3. UNTOLERANCED DIMS +/-0.1 (TURNED), +/-0.2 (MILLED).
2. MAIN SEATS 72 K6 (6306/C3, LINE-BORED WITH BOTH CASE HALVES BOLTED), JOURNALS 30 k5; CRANKPIN BORES 27.915 +/-0.004 (PRESS 0.065-0.082, SLIP SF 2.1),
   CRANKPIN 28 g5 Ra 0.2 (NEEDLE RACE); ROD BIG END 34 H5 CARBURISED 60 HRC Ra 0.2; CRANK TRUED TO 0.02 TIR AT THE MAINS AFTER PRESSING.
3. LINER BORE 62.00 +0.01/0 HONED Ra 0.4 PLATEAU, PISTON CLEARANCE 0.04; LINERS PRESSED 0.05 INTO 68 H7; DECK FLAT 0.03.
4. HEAD: CAM LINE BORES 22.04 +0.02/0 WITH CAPS FITTED (MATCHED, NUMBERED); BUCKET BORES 25.02 +0.02/0; GUIDE BORES 9.5 -0.03;
   VALVE ANGLE 14 DEG +/-0.1 EACH SIDE. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts*.py.

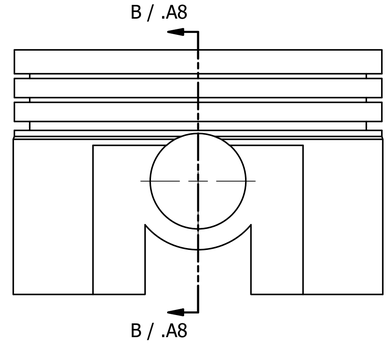
DRAWN Jay Patel	11/2/2026		
CHECKED			
QA		TITLE	TwoCylinderEngine - manufacturing drawings
MFG			
APPROVED		SIZE D	DWG NO TCE-DWG-100 REV
		SCALE 1 : 1	SHEET 1 OF 5

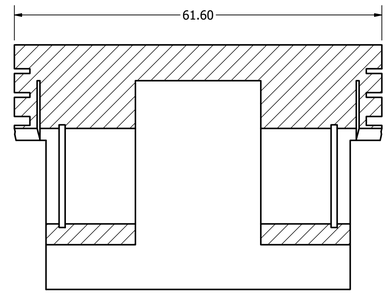
TCE-106 CONROD 4340 QT, BIG END CARBURISED, QTY 2, 232.1 g (INVENTOR), SCALE 2:1, ENVELOPE 131.5 x 46.0 x 18.4 MM
ONE PIECE, CENTRES 94.00 +/-0.02; BIG END 34 H5 x 18.4 (NEEDLE RACE), SMALL END 19 H7 FOR A 16x19 BRONZE BUSH;
I-BEAM 12 THICK, POCKETS 2 DEEP; BORES PARALLEL 0.01 / 100 (QTY 2, MASS-MATCHED +/-1 g)



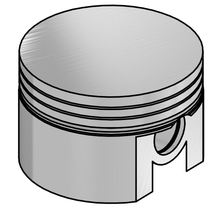
SECTION A-A
SCALE 2 : 1 / .C8

TCE-201 PISTON DONOR PART (EX300 PATTERN) - INSPECTION SHEET, QTY 2, 123.9 g (INVENTOR), SCALE 2:1, ENVELOPE 62.0 x 61.6 x 41.0 MM
BUY: KAWASAKI 13001-0742 (EX300, 62.00): COMPRESSION HEIGHT 22 ASSUMED - MEASURE THE DONOR PISTON AND SET
THE DECK BY THE BASE-GASKET THICKNESS (0.3 MM CROWN-TO-DECK); PIN 16 x 44.5; RINGS 0.8 / 0.8 / 1.5





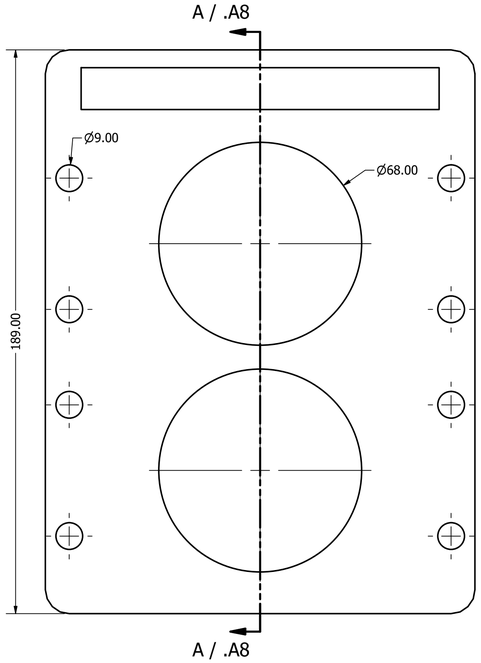
SECTION B-B
SCALE 2 : 1 / .D4

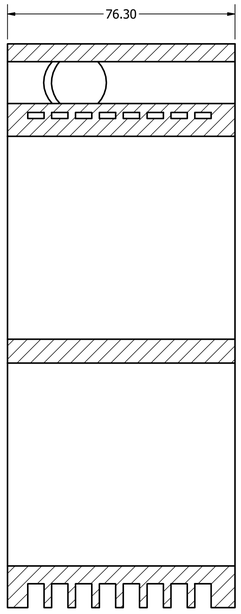


NOTES:
1. 4340 QT 36-40 HRC (CRANK WEBS, CENTRE, RODS); 52100 60 HRC GROUND (CRANKPINS); 6061-T6 BILLET FOR THE ONE-OFF CASE, BLOCK, HEAD (PRODUCTION: A356-T6 SAND CASTING). BREAK SHARP EDGES 0.3. UNTOLERANCED DIMS +/-0.1 (TURNED), +/-0.2 (MILLED).
2. MAIN SEATS 72 K6 (6306/C3, LINE-BORED WITH BOTH CASE HALVES BOLTED), JOURNALS 30 k5; CRANKPIN BORES 27.915 +/-0.004 (PRESS 0.065-0.082, SLIP SF 2.1), CRANKPIN 28 g5 Ra 0.2 (NEEDLE RACE); ROD BIG END 34 H5 CARBURISED 60 HRC Ra 0.2; CRANK TRUED TO 0.02 TIR AT THE MAINS AFTER PRESSING.
3. LINER BORE 62.00 +0.01/0 HONED Ra 0.4 PLATEAU, PISTON CLEARANCE 0.04; LINERS PRESSED 0.05 INTO 68 H7; DECK FLAT 0.03.
4. HEAD: CAM LINE BORES 22.04 +0.02/0 WITH CAPS FITTED (MATCHED, NUMBERED); BUCKET BORES 25.02 +0.02/0; GUIDE BORES 9.5 -0.03; VALVE ANGLE 14 DEG +/-0.1 EACH SIDE. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts*.py.

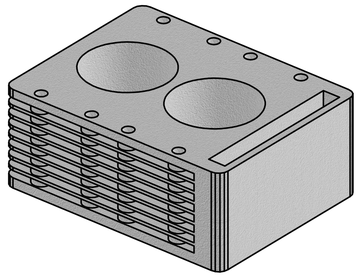
DRAWN Jay Patel	11/2/2026	
CHECKED		TITLE: TwoCylinderEngine - manufacturing drawings
QA		
MFG		
APPROVED		
		SIZE D · DWG NO TCE-DWG-100 · REV
		SCALE 2 : 1 · SHEET 2 OF 5

TCE-403 CYLINDERBLOCK 6061-T6 BILLET (A356-T6 CAST), QTY 1, 2241.1 g (INVENTOR), SCALE 1:1, ENVELOPE 189.0 x 144.0 x 76.3 MM
2 x LINER BORES 68 H7 ON 76.00 +/-0.02 CENTRES; 8 x STUD HOLES 9 AT x = +/-16, +/-60, y = +/-64; FINS 2.5 AT 8 PITCH;
CAM-CHAIN TUNNEL 14 x 120, TENSIONER BORE 20.5; DECK 140.80 FROM THE CRANK AXIS

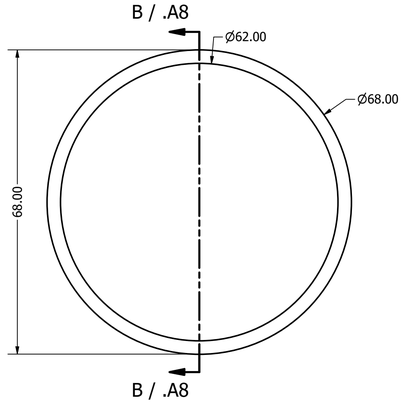


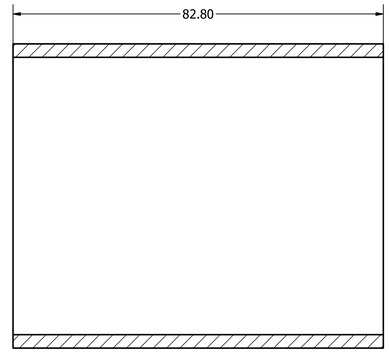


SECTION A-A
SCALE 1 : 1 / .C8



TCE-404 LINER CENTRIFUGAL CAST IRON, QTY 2, 365.2 g (INVENTOR), SCALE 1.5:1, ENVELOPE 82.8 x 68.0 x 68.0 MM
62.00 H6 BORE HONED, 68 OD (PRESS 0.05), 82.8 LONG, SPIGOT 6.5 BELOW THE BLOCK (QTY 2)





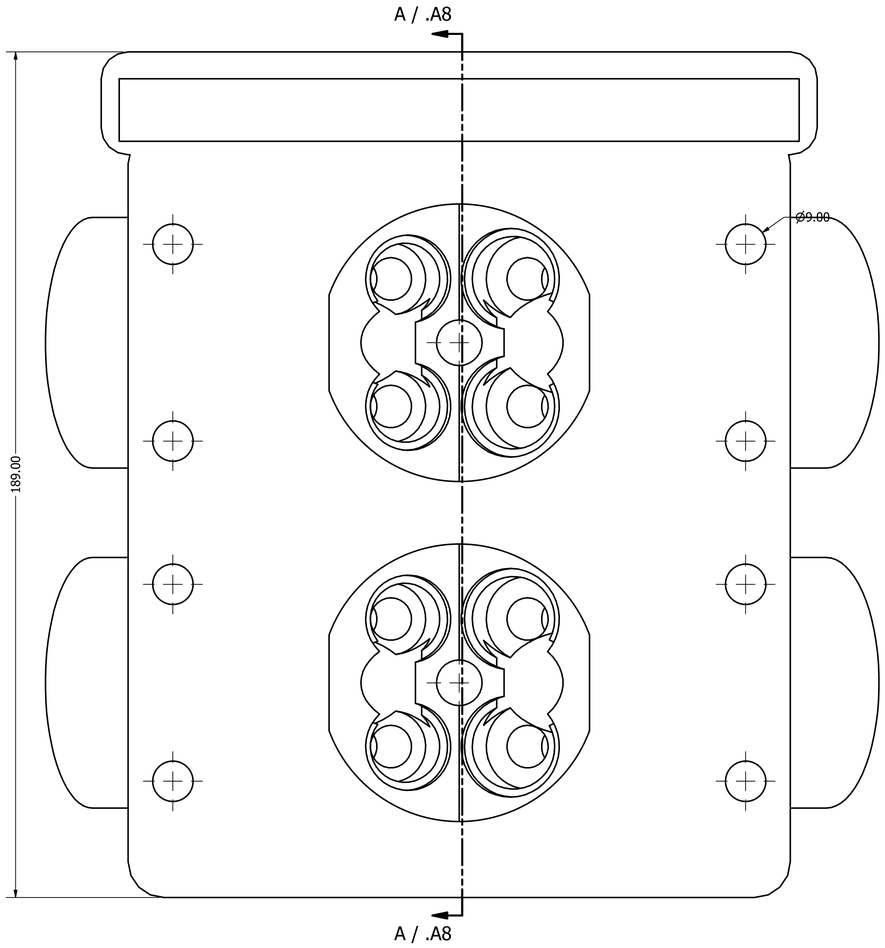
SECTION B-B
SCALE 1.50 : 1 / .D4

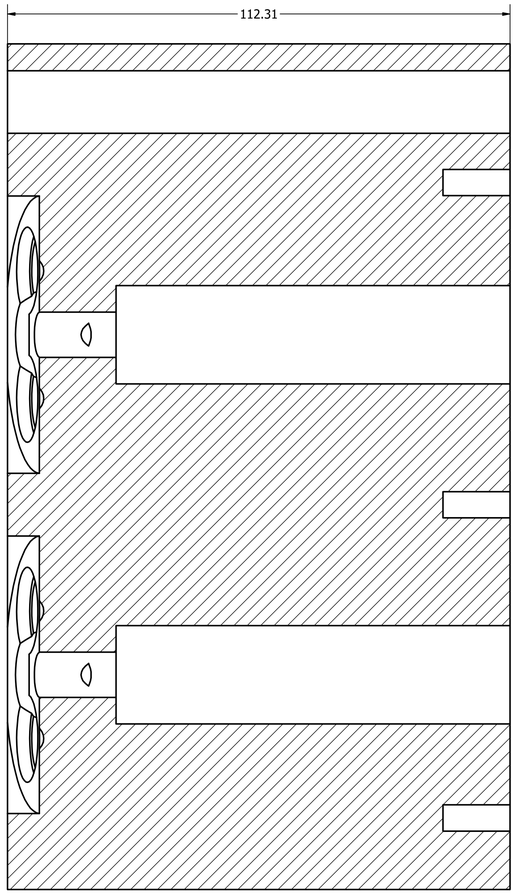


NOTES:
1. 4340 QT 36-40 HRC (CRANK WEBS, CENTRE, RODS); 52100 60 HRC GROUND (CRANKPINS); 6061-T6 BILLET FOR THE ONE-OFF CASE, BLOCK, HEAD (PRODUCTION: A356-T6 SAND CASTING). BREAK SHARP EDGES 0.3. UNTOLERANCED DIMS +/-0.1 (TURNED), +/-0.2 (MILLED).
2. MAIN SEATS 72 K6 (6306/C3, LINE-BORED WITH BOTH CASE HALVES BOLTED), JOURNALS 30 k5; CRANKPIN BORES 27.915 +/-0.004 (PRESS 0.065-0.082, SLIP SF 2.1), CRANKPIN 28 g5 Ra 0.2 (NEEDLE RACE); ROD BIG END 34 H5 CARBURISED 60 HRC Ra 0.2; CRANK TRUED TO 0.02 TIR AT THE MAINS AFTER PRESSING.
3. LINER BORE 62.00 +0.01/0 HONED Ra 0.4 PLATEAU, PISTON CLEARANCE 0.04; LINERS PRESSED 0.05 INTO 68 H7; DECK FLAT 0.03.
4. HEAD: CAM LINE BORES 22.04 +0.02/0 WITH CAPS FITTED (MATCHED, NUMBERED); BUCKET BORES 25.02 +0.02/0; GUIDE BORES 9.5 -0.03; VALVE ANGLE 14 DEG +/-0.1 EACH SIDE. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts*.py.

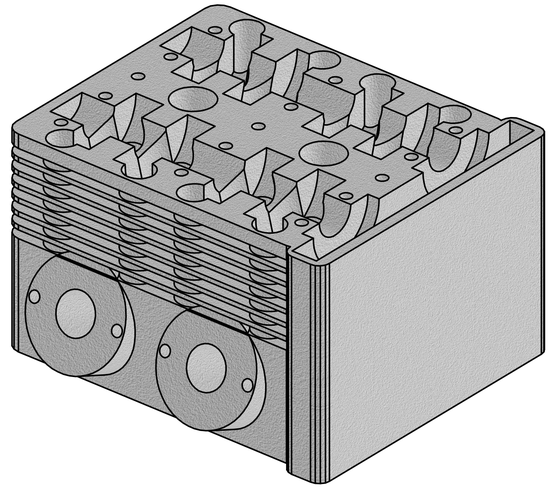
DRAWN Jay Patel	11/2/2026	
CHECKED		TITLE: TwoCylinderEngine - manufacturing drawings
QA		
MFG		
APPROVED		
		SIZE D · DWG NO TCE-DWG-100 · REV
		SCALE 1 : 1 · SHEET 3 OF 5

TCE-405 CYLINDERHEAD 6061-T6 BILLET (A356-T6 CAST), QTY 1, 5983.0 g (INVENTOR), SCALE 1.5:1, ENVELOPE 189.0 x 186.3 x 112.3 MM
PENT ROOF 2 x 14 DEG, RIDGE 7.26 ABOVE THE FACE, CHAMBER 12.01 CC (CR 10.6 WITH 0.8 GASKET);
8 VALVES: INTAKE 22 / EXHAUST 19 ON 28.5 x 22.8 SPACING; BUCKET BORES 25.02; CAM LINES 76.60 APART, 112.31 ABOVE THE FACE





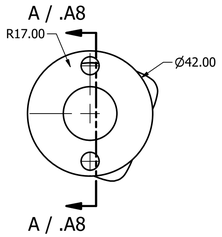
SECTION A-A
SCALE 1.50 : 1 / .C7

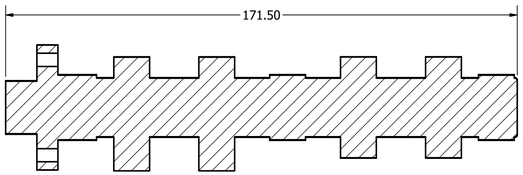


NOTES:
1. 4340 QT 36-40 HRC (CRANK WEBS, CENTRE, RODS); 52100 60 HRC GROUND (CRANKPINS); 6061-T6 BILLET FOR THE ONE-OFF CASE, BLOCK, HEAD (PRODUCTION: A356-T6 SAND CASTING). BREAK SHARP EDGES 0.3. UNTOLERANCED DIMS +/-0.1 (TURNED), +/-0.2 (MILLED).
2. MAIN SEATS 72 K6 (6306/C3, LINE-BORED WITH BOTH CASE HALVES BOLTED), JOURNALS 30 k5; CRANKPIN BORES 27.915 +/-0.004 (PRESS 0.065-0.082, SLIP SF 2.1), CRANKPIN 28 g5 Ra 0.2 (NEEDLE RACE); ROD BIG END 34 H5 CARBURISED 60 HRC Ra 0.2; CRANK TRUED TO 0.02 TIR AT THE MAINS AFTER PRESSING.
3. LINER BORE 62.00 +0.01/0 HONED Ra 0.4 PLATEAU, PISTON CLEARANCE 0.04; LINERS PRESSED 0.05 INTO 68 H7; DECK FLAT 0.03.
4. HEAD: CAM LINE BORES 22.04 +0.02/0 WITH CAPS FITTED (MATCHED, NUMBERED); BUCKET BORES 25.02 +0.02/0; GUIDE BORES 9.5 -0.03; VALVE ANGLE 14 DEG +/-0.1 EACH SIDE. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts*.py.

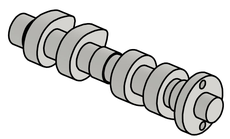
DRAWN Jay Patel	11/2/2026	
CHECKED		TITLE: TwoCylinderEngine - manufacturing drawings
QA		
MFG		
APPROVED		
		SIZE D · DWG NO TCE-DWG-100 · REV
		SCALE 1.50 : 1 · SHEET 4 OF 5

TCE-301 CAMSHAFTINTAKE 4140 QT, NITRIDED, QTY 1, 760.5 g (INVENTOR), SCALE 1:1, ENVELOPE 171.5 x 43.6 x 43.6 MM
4 LOBES: BASE R17, LIFT 7.15, 8TH-ORDER POLYNOMIAL +/-68 CAM DEG (LOBE TABLE FROM design.lobe_profile);
NOSES AT 233 / 323 DEG (EXHAUST 126 / 216); JOURNALS 22 f7 x 3; FLANGE 42 WITH 2 x M6 (SPROCKET)

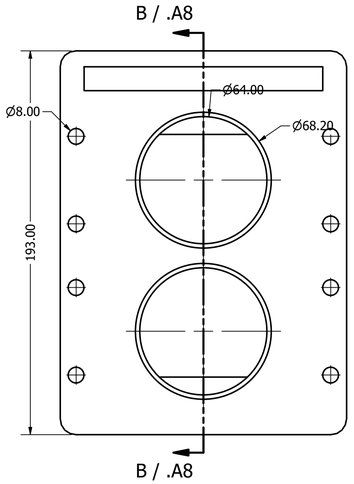


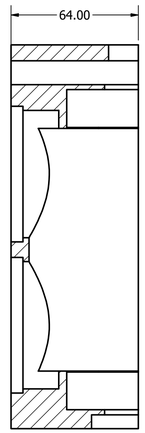


SECTION A-A
SCALE 1 : 1 / .D8

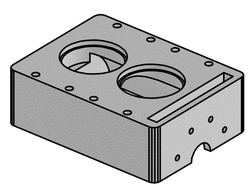


TCE-401 CRANKCASEUPPER 6061-T6 BILLET (A356-T6 CAST), QTY 1, 2441.5 g (INVENTOR), SCALE 1:1.5, ENVELOPE 193.0 x 144.0 x 64.0 MM
HORIZONTALLY SPLIT AT THE CRANK AXIS; 2 x 6306 SEATS 72 K6 LINE-BORED WITH THE LOWER HALF; LINER SPIGOT BORES 68.2;
8 x M8 STUD THREADS, 6 x M8 CASE-BOLT THREADS, CHAIN CAVITY 12 x 120

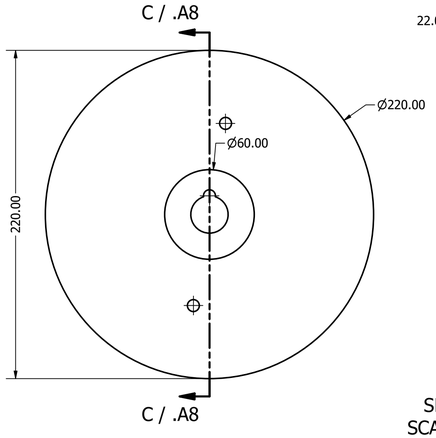


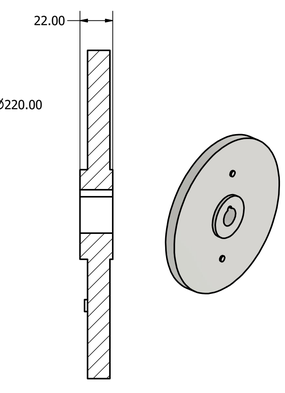


SECTION B-B
SCALE 2 / 3 / .D4



TCE-108 FLYWHEEL 1045, QTY 1, 4543.6 g (INVENTOR), SCALE 1:2, ENVELOPE 220.0 x 220.0 x 22.0 MM
1045: DISC 220 x 15 (RING-GEAR SHRINK SEAT 220 s6), HUB 60 x 22 BORE 25 H7 + ROUND KEY 8; 2 TRIGGER TABS AT r 62, 180 DEG
(RING GEAR KOHLER 97T 10 DP SHRUNK ON AT 200 C); BALANCE G6.3





SECTION C-C
SCALE 1 / 2 / .B8

NOTES:
1. 4340 QT 36-40 HRC (CRANK WEBS, CENTRE, RODS); 52100 60 HRC GROUND (CRANKPINS); 6061-T6 BILLET FOR THE ONE-OFF CASE, BLOCK, HEAD (PRODUCTION: A356-T6 SAND CASTING). BREAK SHARP EDGES 0.3. UNTOLERANCED DIMS +/-0.1 (TURNED), +/-0.2 (MILLED).
2. MAIN SEATS 72 K6 (6306/C3, LINE-BORED WITH BOTH CASE HALVES BOLTED), JOURNALS 30 k5; CRANKPIN BORES 27.915 +/-0.004 (PRESS 0.065-0.082, SLIP SF 2.1), CRANKPIN 28 g5 Ra 0.2 (NEEDLE RACE); ROD BIG END 34 H5 CARBURISED 60 HRC Ra 0.2; CRANK TRUED TO 0.02 TIR AT THE MAINS AFTER PRESSING.
3. LINER BORE 62.00 +0.01/0 HONED Ra 0.4 PLATEAU, PISTON CLEARANCE 0.04; LINERS PRESSED 0.05 INTO 68 H7; DECK FLAT 0.03.
4. HEAD: CAM LINE BORES 22.04 +0.02/0 WITH CAPS FITTED (MATCHED, NUMBERED); BUCKET BORES 25.02 +0.02/0; GUIDE BORES 9.5 -0.03; VALVE ANGLE 14 DEG +/-0.1 EACH SIDE. ALL DIMENSIONS MM. SOURCE: _build/design.py + build_parts*.py.

DRAWN Jay Patel	11/2/2026		
CHECKED			
QA			
MFG			
APPROVED			
		SIZE D	DWG NO TCE-DWG-100 REV
		SCALE 1 : 1	SHEET 5 OF 5

TITLE: TwoCylinderEngine - manufacturing drawings